

A

Fixed spectrum; different interactions

● 105 perfect matchings M of $x_1 \dots x_8$

$$f_M = \frac{1}{2} \sum_{(i,j) \in M} x_i x_j$$

▲ 35 four-plus-four partitions $A \mid A'$

$$f_A = \frac{1}{2} (\prod_{i \in A} x_i + \prod_{i \in A'} x_i)$$

■ 24 nested-prefix controls (D–F)

$\frac{1}{2} \times$ products of the first 2, 4, 6, 8 inputs of a seeded permutation

matching, quartic: input-gradient

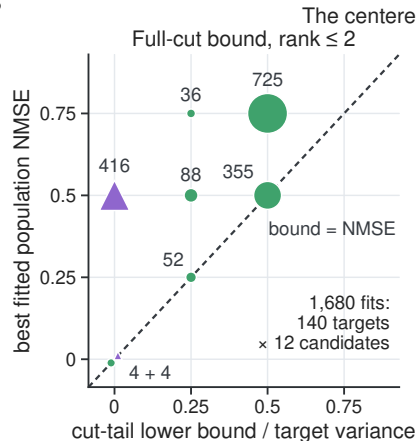
second moment $I_8 / 4$, rank 8

nested: anisotropic spectrum

$(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}, \frac{1}{2}, \frac{3}{4}, \frac{3}{4}, 1, 1)$

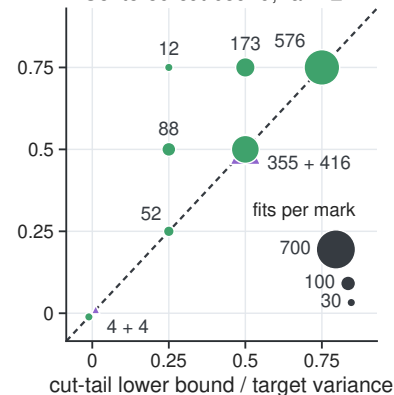
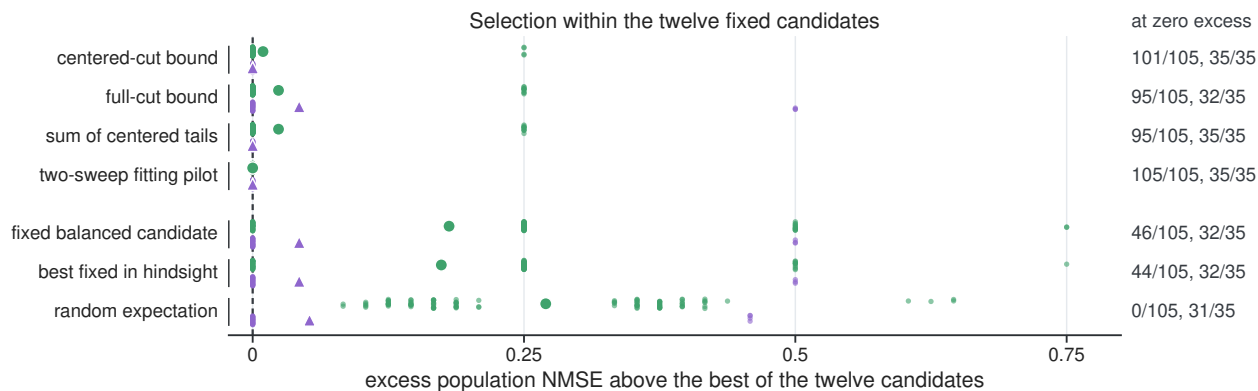
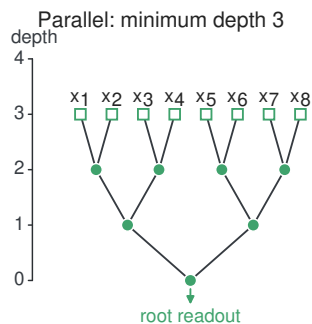
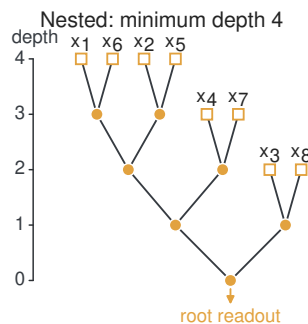
every tree: 7 multi-affine nodes,

28 coefficients, 14 edges, root readout

B

The centered constraint is stronger

Centered-cut bound, rank ≤ 1

**C****D****E****F**